

MATERIAL SAFETY DATA SHEET

Product Name:	SPU 6-93R	File Name:	msds-6-93R-5
Date of preparation: 10/01/2014	Issue No: 5	Page :	1 of 3

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION			
Product Name	SPU 6-93R	Use : Zinc phosphating solution	
Manufacturer's Name	: UNASCO (M) SDN.BHD.		
Address	: No.1, Jln PJU 1A/18, Tmn Perindustrian Jaya Subang, 46050 Petaling Jaya, Selangor.	Tel. No : 03-7847 3468	
Emergency Tel. No.	: 03-7847 3468		
Contact Point	: Technical Manager	Tel. No. : 03-7847 3468	
SECTION 2 : COMPOSITION/INFORMATION OF INGREDIENT			
Chemical Name	CAS No.	Proportion	Exposure Limit
Phosphoric acid	7664-38-2	>10%	1 mg/m ³
Nitric acid	7697-37-2	>5%	2 ppm
SECTION 3 : PHYSICAL AND CHEMICAL PROPERTIES			
Appearance	Green liquid		
Solubility in water	Complete	Melting Point	N/A °C
Boiling Point	N/A °C	Flash Point	N/A °C
Auto-ignition temperature	N/A °C	Flammable Limit	Non-flammable
Specific Gravity	1.516	pH Value	2.5±0.5 (1% w/v solution)
SECTION 4 : HAZARD IDENTIFICATION			
Most important hazard : Contact with combustible material may cause fire. Cause severe burns. Risk of serious damage to eyes.			
SECTION 5 : FIRST AID MEASURE			
Ingestion	Do not induce vomiting. Give two glasses of water and seek medical attention immediately.		
Eye contact	Immediately flush with running water for at least 15 minutes. Seek medical attention without delay.		
Skin contact	Immediately remove contaminated clothing and shoes-launders before re-use. Wash skin thoroughly with water. Seek medical advice if redness and irritation occurs.		
Inhalation	Remove to fresh air. Give artificial respiration if not breathing. Obtain medical attention.		

MATERIAL SAFETY DATA SHEET

Product Name:	SPU 6-93R	File Name:	msds-6-93R-5
Date of preparation: 10/01/2014	Issue No: 5	Page :	2 of 3

SECTION 6 : FIRE FIGHTING MEASURE	
Extinguishing Media	Select extinguishing media suitable for surrounding fire.
Fire fighting instruction	Wear self-contained breathing apparatus and protective clothing. Cool fire exposed container with water spray.
Special Hazard	Oxidizer. Contact with combustible material, flammable material, or powder metals can cause fire or explosion. Reacts with most metals to release hydrogen gas.
SECTION 7 : ACCIDENTAL RELEASE MEASURE	
Leak and / or Spill	Ventilate area of leak. Wear appropriate personal protective clothing. Contain spillage by damming or construction of barriers. Neutralize with alkali (e.g. Soda ash), then absorb with earth or non-organic absorbent material and place into dry well-labeled container for disposal. Wash spillage area with plenty of water.
SECTION 8 : HANDLING AND STORAGE	
Handling	Open slowly to release potential pressure before removing cap. When diluting, the acid should always be added slowly to water and in small amounts. Avoid contact with skin and eyes. Wash thoroughly after handling.
Storage	Store in a cool, ventilated storage area. Keep out of direct sunlight and away from heat, water and incompatible materials.
SECTION 9 : EXPOSURE CONTROL AND PERSONAL PROTECTION	
a. Exposure Limit	ACGIH TLV : Nitric acid : 2.0 ppm (TWA) Phosphoric acid: 1 mg/m ³ (TWA)
b. Ventilation	A system of local and/or general exhaust is recommended to keep employee exposures below the Airborne Exposure Limits.
c. Personal protection	
Eye	Use chemical safety goggles and/or a full face shield where splashing is possible. Maintain eye wash fountain and quick-drench facilities in work area.
Skin	Wear impervious protective clothing, including boots, gloves, apron or coveralls, as appropriate, to prevent skin contact.
Respiratory protection	If the exposure limit is exceeded, wear a NIOSH approved respirator.
SECTION 10: STABILITY AND REACTIVITY	
Condition to avoid	Stable under ordinary conditions of use and storage. Avoid light and heat.
Incompatibles	Strong bases, metallic powder, carbides and combustible organics.
Decomposition products	N/A
Hazardous Polymerization	Will not occur.
SECTION 11 : ECOLOGICAL INFORMATION	
Mobility	Soluble in water.
Bioaccumulation	N/A
Biodegradability	N/A
Aquatic Toxicity	This material is expected to be toxic to aquatic life.

MATERIAL SAFETY DATA SHEET

Product Name:	SPU 6-93R	File Name:	msds-6-93R-5
Date of preparation: 10/01/2014	Issue No: 5	Page :	3 of 3

SECTION 12 : TOXICOLOGICAL INFORMATION

Toxicity Data	Nitric acid Phosphoric acid	LC ₅₀ inhalation (rat) :244 ppm(NO ₂)/30M Oral LD ₅₀ (rat) : 1530 mg/kg										
Carcinogenicity	N/A											
Effects of overexposure												
<table><tr><td>Route of entry</td><td>Effects</td></tr><tr><td>Inhalation</td><td>Corrosive.</td></tr><tr><td>Skin contact</td><td>Corrosive. Can cause redness, pain and severe skin burns.</td></tr><tr><td>Eye contact</td><td>Corrosive. Contact may cause severe burns and permanent eye damage.</td></tr><tr><td>Ingestion</td><td>Corrosive. Swallowing can cause immediate pain and burns of mouth, throat, esophagus and gastrointestinal tract.</td></tr></table>			Route of entry	Effects	Inhalation	Corrosive.	Skin contact	Corrosive. Can cause redness, pain and severe skin burns.	Eye contact	Corrosive. Contact may cause severe burns and permanent eye damage.	Ingestion	Corrosive. Swallowing can cause immediate pain and burns of mouth, throat, esophagus and gastrointestinal tract.
Route of entry	Effects											
Inhalation	Corrosive.											
Skin contact	Corrosive. Can cause redness, pain and severe skin burns.											
Eye contact	Corrosive. Contact may cause severe burns and permanent eye damage.											
Ingestion	Corrosive. Swallowing can cause immediate pain and burns of mouth, throat, esophagus and gastrointestinal tract.											
Chronic Effects		N/A										
Target Organs		N/A										
Medical conditions generally aggravated by exposure		N/A										

SECTION 13 : DISPOSAL INFORMATION

Disposal of waste to be in accordance with the Environmental Quality (Scheduled Wastes) Regulations 2005 and other guidelines issued by DOE and /or local authorities.

SECTION 14 : TRANSPORT INFORMATION

UN/NA Number	UN3264
T.D.G classification	Class 8 – Corrosive (Liquid)
Packing group	II
Special Shipping instruction	N/A

SECTION 15 : REGULATORY INFORMATION

a. Proposed classification :			Corrosive
b. Risk phrase :	R8	Contact with combustible material may cause fire.	
	R41	Risk of serious damage to eyes.	
	R35	Cause severe burns.	
c. Safety phrase :	S23	Do not breathe fume.	
	S26	In case of contact with eyes, rinse immediately with large amount of water and seek medical advice.	
	S36	Wear suitable protective clothing.	

SECTION 16: OTHERS INFORMATION

The data given here is based on our current knowledge and experience. The purpose of this Safety Data sheet is to describe the product in terms of their health and safety requirements. Since the conditions of use are outside our UNASCO's control, any recommendations or suggestions are made without guarantee and UNASCO disclaims any liability for loss or damage suffered from use of this information. User is responsible for safe working.